MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2MDCECT100 - IoT based smart farming

Duration: 1½ hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- Demonstrate the given task(Any one of the tasks listed below)
- Viva session (Minimum 5 questions from the Practical Module)
- Lab Report/ Case study & field visit Report

Evaluation Criteria:

- Demonstration (Lab/ Presentation: 15 marks)
- Case Study& field visit report/ Lab report: 10 marks
- Viva: 10 marks

CHOOSE ANY ONE

1. Prepare a case study report /field visit report based on a simple hydroponic system using readily available materials.
2. Conduct a case study report/field visit report growth rate of plants in soil and a hydroponic system over two weeks and document your observations.
3. Furnish a case study report/field visit report on data collection from a soil moisture sensor in an agriculture farm.
4. Prepare a case study report/field visit report on automated drip irrigation using a soil moisture sensordata.
5. Conduct a case study report/field visit report crop monitoring system using a web cam / ESP 32 cam in a controlled environment and report the findings.
6. Prepare a case study report/ field visit report on any three advancedtechnologies used in the smart farm you visited.
7. Furnish a case study report/field visit report on the productivity and efficiency of a smart farm versus a conventional farm.
8. Conduct a case study report/field visit report on key advantages and challenges of implementing vertical farming in urban areas compared to traditional farming.
9. Prepare a case study report/field visit report on aquaponics integrating plant and fish farming to create a self-sustaining system, and what are the economic implications?
10. Furnish a case study report/field visit report on IoT-based UV bug trap function, and what are its advantages over conventional pest control methods?

11. Conduct a case study report/ field visit report on the challenges in implementing IoT-based UV bug traps in large-scale farming operations?
12. Write an Arduino program to blink an LED at a 1-second interval.
13. Develop an Arduino program to adjust the LED blink rate to 100 milliseconds on time and 1000 milliseconds off time.
14. Write an Arduino program to create an LED chaser effect with 6 LED.
15. Write an Arduino chaser program to reverse the direction of movement every 5 seconds.
16. Develop an Arduino program to blink an LED on when a push button is pressed and off when released.
17. Write an Arduino program to fade an LED state with a button press.
18. Write an Arduino program to blink an LED 15 times with a 1-second delay.
19. Write an Arduino program to cycle through red, green, and blue colors on an RGB LED.
20. Develop an Arduino program to mix colours and display purple, yellow, and cyan using RGB LED.
21. Write an ESP 32 program to increase the LED blink rate when a button is pressed remotely.
22. Write an ESP 32 program to toggle the LED state with a button press.
23. Write an ESP 32 program to simulate a basic traffic light system using an RGB LED remotely.
24. Develop an ESP 32 program to simulate a basic traffic light system using an RGB LED, including a pedestrian crossing delay by a wait button.
25. Write an ESP 32 program to send a notification on when an object is detected by an IR sensor.
26. Write an Arduino program to turn off the LED automatically after 5 seconds if no object is detected.